

Federated Learning Approach for Obstructive Sleep Apnea Prediction

Bachelor thesis presentation

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Outline

Introduction

Project Background

Objective within the Bachelor Project

Research and/or Development

Solution

Project Approach

Output Demo

Reflection

Introduction

- ▶ Team
 - ▶ Manja Gorenc Novak, *Head of AI and Senior Data Scientist*
 - ▶ Rok Vinder, *Data Analyst and Machine Learning Engineer*
- ▶ Result d.o.o
 - ▶ Slovenian Company
 - ▶ ~ 50 employees
 - ▶ Software development
 - ▶ Data and AI
 - ▶ Business solutions
- ▶ Context
 - ▶ eEarly
 - ▶ Health data
 - ▶ Obstructive sleep apnea

Project background

Problem statement

- ▶ Problem statement
 - ▶ Machine learning
 - ▶ Centralized
 - ▶ Private (health) data
 - ▶ Regulations

Project background

Stakeholders and their needs

- ▶ **Developers**

- ▶ Understand code
- ▶ Interpret results
- ▶ Run the project

- ▶ *Patients*

- ▶ **Data privacy**

- ▶ *Doctors*

- ▶ *Improved diagnosis*
- ▶ *Better tools*

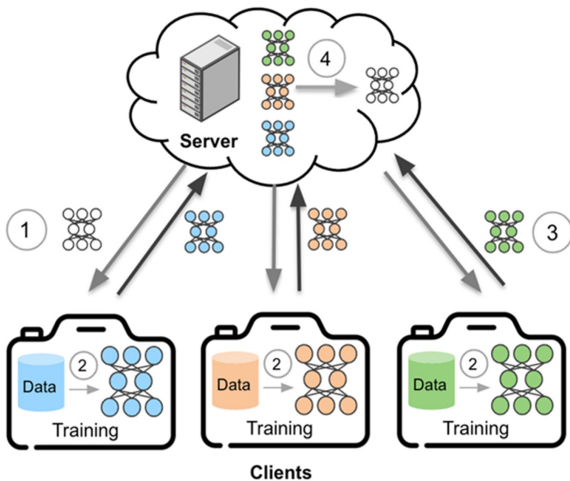
Objective within the Bachelor Thesis

Proposed solution

- ▶ Proposed solution
 - ▶ Federated learning
 - ▶ ~~Data to the model (centralized)~~
 - ▶ Model to the data (decentralized)
 - ▶ Privacy, Regulations
 - ▶ Bandwidth

Objective within the Bachelor Thesis

Federated learning algorithm



Objective within the Bachelor Thesis

Scope

Table: Red color represents MVP, orange and blue represent broader or ultimate scope.

Importance	Requirement
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning (POC).
Should have	Single-machine and containerized (with Docker) – –
Could have	Multi-machine and containerized (with Docker) – –
Could have	Multi-machine and containerized (on Kubernetes) – –
Could have	Differential privacy mechanism implemented.

Research and/or development

Comparative study and technology decision

Flower

- ▶ beginner friendly
- ▶ prototype friendly
- ▶ good documentation
- ▶ excellent integration (!)

NVIDIA FLARE

- ▶ enterprise ready
- ▶ many built-in features
- ▶ NVIDIA ecosystem
- ▶ heavy for small projects

TensorFlow Federated

- ▶ widely used
- ▶ complex
- ▶ the least integration

Solution

Final solution

- ▶ Federated learning network
- ▶ Single-machine (non-containerized)
- ▶ Multi-machine (containerized)

Project Approach

Development method

Scrumban

- ▶ Daily meetings
- ▶ Agile mindset
- ▶ Backlog
- ▶ Typical kanban board

Project Approach

Work division and Collaboration

Work division

- ▶ MVP
 - ▶ Flower quickstart examples
 - ▶ Connecting with existing apnea prediction model
 - ▶ Preparing the data
- ▶ Additional features
 - ▶ Containerizing with Docker
 - ▶ Connecting multiple machines

Collaboaration

- ▶ Daily meetings
- ▶ Slack and Outlook

Output and Demo

Reflection

Achieved objectives

Table: Red color represents MVP, orange and blue represent broader or ultimate scope.

Importance	Requirement	Done
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning.	Yes
Should have	Single-machine and containerized (with Docker) – –	Yes
Could have	Multi-machine and containerized (with Docker) – –	Yes
Could have	Multi-machine and containerized (on Kubernetes) – –	No
Could have	Differential privacy mechanism implemented.	No

Reflection

Added value and impact

- ▶ Privacy (not only patients)
- ▶ Better diagnosis tools (doctors)
- ▶ Regulations (developers)
- ▶ Less network usage (climate)
- ▶ Project opportunities (Result)
- ▶ Research (scientists)

Reflection

Improvements, and choice reflection

Improvements and weaknesses

- ▶ Evaluation metrics
- ▶ Separate networks
- ▶ Real TLS certification

Choice reflection

- ▶ Excellent
- ▶ No change

Thank you for your attention!

Questions and Answers